CEUE203 - OOP25AIML034

PRACTICAL – 3

Aim:

1. Distinct Points: To create a Point class in Java and override toString(), equals(), and hashCode() methods to identify and count distinct points in an array.

2. Duplicate Card Check: To create a Card class in Java and override toString(), equals(), and hashCode() methods to detect duplicate cards in an array.

3. Fractions (Advanced Learners): To create a Fraction class in Java that reduces fractions to their lowest terms and overrides toString(), equals(), and hashCode() methods to compare equivalent fractions.

Code Screenshot:

```
import java.util.Objects;


class Point {
    private int x, y;


    public Point(int x, int y) {
        this.x = x;
        this.y = y;
    }


    @Override
```

```
    public String toString() {
        return "(" + x + ", " + y + ")";
    }

    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (!(obj instanceof Point)) return false;

        Point p = (Point) obj;
        return x == p.x && y == p.y;
    }

    @Override
    public int hashCode() {
        return Objects.hash(x, y);
    }
}

public class Main {
    public static void main(String[] args) {
        Point[] points = {
```

```
    new Point(1, 2),
    new Point(3, 4),
    new Point(1, 2),
    new Point(5, 6),
    new Point(3, 4)
};

int distinct = 0;

for (int i = 0; i < points.length; i++) {
    boolean found = false;

    for (int j = 0; j < i; j++) {
        if (points[i].equals(points[j])) {
            found = true;
            break;
        }
    }

    if (!found) {
        distinct++;
    }
```

```
        }

        System.out.println("Distinct: " + distinct);
    }
}
```

2.

```
import java.util.Objects;

class Card {
    private String rank, suit;

    public Card(String rank, String suit) {
        this.rank = rank;
        this.suit = suit;
    }

    @Override
    public String toString() {
        return rank + " of " + suit;
    }
```

```
    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (!(obj instanceof Card)) return false;

        Card c = (Card) obj;
        return Objects.equals(rank, c.rank)
            && Objects.equals(suit, c.suit);
    }

    @Override
    public int hashCode() {
        return Objects.hash(rank, suit);
    }
}

public class Main {
    public static void main(String[] args) {
        Card[] cards = new Card[5];

        cards[0] = new Card("King", "Hearts");
        cards[1] = new Card("Ace", "Spades");
```

```
int count = 2;


Card[] newCards = {
   new Card("Queen", "Diamonds"),
   new Card("Ace", "Spades"),
   new Card("Jack", "Clubs")
};


for (Card card : newCards) {
   boolean duplicate = false;


   for (int i = 0; i < count; i++) {
      if (card.equals(cards[i])) {
         System.out.println("Duplicate found: " + card);
         duplicate = true;
         break;
      }
   }


   if (duplicate) {
      break;
```

```
            }

            cards[count] = card;
            count++;
        }
    }
}
```

3.

```
import java.util.Objects;

class Fraction {
    private int num, den;

    public Fraction(int num, int den) {
        if (den == 0) {
            throw new IllegalArgumentException(
                "Denominator cannot be zero"
            );
        }

        if (den < 0) {
```

```
        num = -num;
        den = -den;
    }

    int g = gcd(Math.abs(num), den);
    this.num = num / g;
    this.den = den / g;
}

private static int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

@Override
public String toString() {
    return num + "/" + den;
}
```

```
    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (!(obj instanceof Fraction)) return false;

        Fraction f = (Fraction) obj;
        return num == f.num && den == f.den;
    }

    @Override
    public int hashCode() {
        return Objects.hash(num, den);
    }
}

public class Main {
    public static void main(String[] args) {
        Fraction f1 = new Fraction(1, 2);
        Fraction f2 = new Fraction(2, 4);
        Fraction f3 = new Fraction(3, 6);
```

```
        System.out.println("f1 = " + f1);

        System.out.println("f2 = " + f2);

        System.out.println("f3 = " + f3);



        System.out.println("f1 equals f2: " + f1.equals(f2));

        System.out.println("f2 equals f3: " + f2.equals(f3));

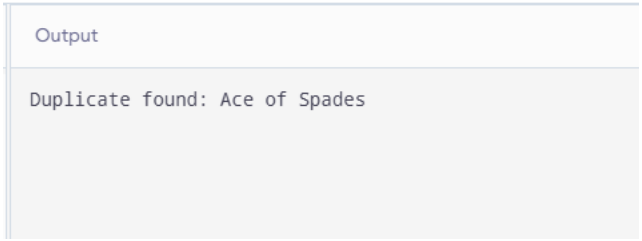
        System.out.println("f1 equals f3: " + f1.equals(f3));

    }

}
```

Output Screenshot:





Output

```
f1 = 1/2
f2 = 1/2
f3 = 1/2
f1 equals f2: true
f2 equals f3: true
f1 equals f3: true
```